

EMPLOYEE TURNOVER PREDICTIVE ANALYTICS

Using workforce data to support targeted retention decisions

Selected Portfolio Sample
People Analytics & Predictive Modeling

Adapted from an independent graduate Advanced Analytics project using an assigned, simulated human resources dataset. This sample highlights the analytical workflow, model comparison, decision-threshold analysis, and translation of predictive results into a business recommendation.

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1. Business Problem & Dataset

The project examined whether employee turnover could be predicted using available workforce data and how those predictions could be translated into practical retention decisions. The goal was not simply to maximize a model metric, but to identify a model and decision rule that could support targeted action while limiting unnecessary interventions.

9,999	23.6%	70 / 30	9
employee records simulated dataset	employees who left 2,364 of 9,999	stratified train / test split	predictors retained after Boruta

The stratified split preserved the turnover distribution: the training set contained 6,998 employees (1,654 leavers; 23.6%) and the held-out test set contained 3,001 employees (710 leavers; 23.7%).

Exploratory analysis showed meaningful relationships between turnover and several aspects of employee experience, especially satisfaction and workload. These patterns were treated as predictive signals, not proof of causation.

Core Business Questions

Prediction Can workforce data distinguish employees who leave from those who stay?	Decision Which model and threshold best balance risk identification with unnecessary action?	Application How can predictions inform targeted retention without replacing human judgment?
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2. Analytical Workflow & Data Integrity

The analysis used an end-to-end modeling workflow designed to keep the held-out test set independent until final evaluation. Feature selection and model development were conducted using training data, with 10-fold stratified cross-validation used for tuning and comparison.

Data Preparation	Exploratory Analysis	Stratified Split	Boruta Feature Selection
10-Fold Cross-Validation	Model Tuning	Held-Out Evaluation	Threshold + Cost Analysis

Feature Selection

Boruta was run on the training data only. It confirmed all nine predictors as useful. Satisfaction level demonstrated the highest average importance, followed by number of projects, time spent at the company, last evaluation, and average monthly hours.

Protecting the Analysis

No test-set leakage The test set stayed separate during feature selection, tuning, and model comparison.	Consistent resampling Candidate workflows used the same stratified cross-validation framework.	Multiple metrics J-Index, ROC AUC, PR AUC, sensitivity, specificity, and confusion matrices were considered.
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Alternative class-imbalance strategies were also compared, including no sampling, upsampling, downsampling, SMOTE, ADASYN, and class weighting where supported.

3. Model Comparison

Four model families were developed: penalized logistic regression, Random Forest, XGBoost, and Support Vector Machine. Cross-validated J-Index was the primary selection metric, with Precision-Recall AUC and other measures used to test the consistency of the rankings.

Model Family	Best Preprocessing	CV J-Index
Penalized Logistic Regression	SMOTE	0.5876
Random Forest	No Sampling	0.9439
Support Vector Machine	SMOTE	0.8864
XGBoost	No Sampling	0.9453

XGBoost produced the highest cross-validated J-Index, with Random Forest only 0.0014 behind. That result did not automatically determine the final model. The strongest workflow from each family was finalized and evaluated on the completely unseen test set.

Held-Out Test Result

681 of 710 employees who left were correctly identified by Random Forest at the default 0.50 threshold.	3 false positives Only three employees who stayed were incorrectly flagged at the default threshold.	ROC AUC = 0.993 Random Forest maintained excellent threshold-independent discrimination.
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Random Forest was selected for deployment because it generalized extremely well while minimizing incorrect flags - a practical concern because every false positive could translate into an unnecessary retention intervention.

4. From Model Performance to a Decision Threshold

A predictive model outputs probabilities; the organization still has to decide when a probability is high enough to act. The conventional 0.50 cutoff is not automatically the best threshold for a turnover decision.

Statistical Threshold

~0.43 Random Forest maximized J-Index at approximately 0.43-0.44, producing the strongest statistical balance between sensitivity and specificity.	0.50 The conventional default cutoff performed well, but it was not treated as optimal simply because it is standard.
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Financial Threshold

The project then applied the assignment's financial assumptions across alternative thresholds. Once intervention costs, expected intervention success, and avoided turnover costs were considered, the threshold that maximized expected financial return was 0.30.

\$100,000 assumed cost of one employee departure	\$5,000 assumed cost of one targeted retention intervention	50% assumed probability an intervention prevents turnover
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The statistically optimal threshold and the financially optimal threshold answered different questions. Deployment required translating prediction quality into the economics of the decision.

5. Expected Financial Impact Under Project Assumptions

At the financially optimized 0.30 threshold, the Random Forest model produced the highest estimated net savings under the assignment's assumptions.

\$30.74M	702	342.5	17
estimated net savings	employees receiving interventions	expected departures prevented	false-positive interventions

Decision Component	Estimated Result
Employees flagged who would otherwise leave	685
Employees flagged who would otherwise stay	17
Expected departures avoided at 50% success	342.5
Avoided turnover costs	\$34.25 million
Intervention costs	\$3.51 million
Estimated net savings	\$30.74 million

Important: These figures are scenario estimates from a simulated dataset using financial assumptions supplied for the graduate assignment. They demonstrate cost-sensitive decision analysis and are not actual organizational savings.

The business value of the exercise was the process of asking how prediction errors, intervention costs, and expected benefits should influence the point at which an organization chooses to act.

6. Workforce Insights & Responsible Application

Variable-importance analyses consistently highlighted employee experience and workload variables. Random Forest placed the greatest emphasis on satisfaction level, number of projects, average monthly hours, last evaluation, and time spent at the company.

Most Influential Predictive Signals

Satisfaction Strongest and most consistent predictive signal.	Workload Projects and monthly hours contributed strongly.	Tenure Time at the company was consistently useful.	Performance context Last evaluation contributed meaningfully.
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Predictive Does Not Mean Causal

Variable importance reflects contribution to accurate classification. It does not establish that changing a variable will cause turnover to rise or fall. With real organizational data, these signals should guide deeper investigation rather than become automatic prescriptions.

From Risk Score to Human Decision

The final model was positioned as a decision-support tool, not a replacement for managerial judgment. Higher-risk employees could be prioritized for individualized conversations and support - such as stay interviews, career-development discussions, workload adjustments, mentoring, or other manager-led retention efforts - while leaders still determine what is appropriate for the individual employee.

A predictive model can tell an organization where to look. Understanding why a pattern exists requires organizational context and employee insight.

7. Practitioner Reflection

What This Project Demonstrates

This was an independent graduate project completed after entering the course with relatively limited R experience. The work leading up to the final project required substantial practice debugging code, understanding modeling workflows, and learning how small technical choices can affect the integrity of an analysis. By the final project, I was able to independently execute and interpret an end-to-end predictive modeling workflow.

One of the most valuable parts of the project was model comparison. Rather than treating the most complex model - or the model with the single highest cross-validation metric - as automatically superior, I learned to evaluate performance across multiple metrics, unseen data, classification errors, and the practical cost of acting on predictions.

What I Would Add With Real Organizational Data

With real workforce data, I would begin with deeper organizational discovery. Before interpreting predictive patterns, I would want to understand the people behind the numbers: relevant organizational history, workforce practices, employee experiences, changes in leadership or structure, and other context that could shape the data.

My Role

Independent Analyst. I completed the full workflow: data preparation and exploration; train/test design; Boruta feature selection; cross-validation; preprocessing and class-imbalance comparisons; model development and tuning; held-out evaluation; model selection; threshold optimization; financial cost-benefit analysis; variable-importance interpretation; model serialization/reload validation; and technical and manager-facing recommendations.

Portfolio Note

This selected sample is adapted from a larger graduate Advanced Analytics project using an assigned, simulated HR dataset. The repository also includes the underlying Quarto/R analysis for technical review.